Software Design Document (SDD)

System Architecture

The application follows a standard **Client-Server Architecture**.
Frontend (Client): HTML5, CSS3, and Vanilla JavaScript. Handles all UI rendering, form validation, and interactive state changes (e.g., toggling donation types).
Backend (Server): PHP 8.x. Handles API requests, business logic, and session management.
Database: MySQL. Stores persistent data securely.

User Interface (UI) Design

Design Language: Modern, card-based interface with a primary color palette of deep green (`#1f7a4c`) to symbolize growth, health, and hope.
Key Components:

- **Sticky Navbar**: Green background with white text, featuring dynamic "Login" vs "My Profile" states.
- **Fluid Grids**: Used for the "Programs" and "Impact" sections to adapt to screen sizes seamlessly.
- **Interactive Buttons**: Pill-shaped buttons with hover states and active-state styling for donation amounts.

Database Schema (MySQL)

Table 1: `users/donors`
`id` (INT, Primary Key, Auto-Increment)
`name` (VARCHAR)
`email` (VARCHAR, Unique)
`password` (VARCHAR, Hashed)
`idno` (VARCHAR)
‘country’ (VARCHAR)
‘city’ (VARCHAR)
`phone` (VARCHAR)
‘created_at’ (TIMESTAMP)

Table 2: `donations
`id` (INT, Primary Key, Auto-Increment)
`user_id` (INT, Foreign Key referencing `users.id`, nullable for guests)
`program` (VARCHAR - e.g., 'education', 'clean_water')
`plan` (VARCHAR - e.g., 'one-time', 'monthly')
`amount` (DECIMAL)
`payment_method` (VARCHAR - e.g., 'paypal', 'mpesa')
‘status’ (VARCHAR, DEFAULT- pending)
‘transaction_id’ (VARCHAR)
‘donation_date’ (TIMESTAMP)
`created_at` (TIMESTAMP)/’updated_at’

Table 3: ‘messages/inquiries’

‘ message_id’ (INT, Primary Key, Auto-Increment)
`name` (VARCHAR)
`email` (VARCHAR, Unique)
‘subject’ (VARCHAR),
‘Message’ (VARCHAR)
‘status’ (VARCHAR DEFAULT- 'unread'),
‘created_at’ (TIMESTAMP DEFAULT -CURRENT_TIMESTAMP),
replied_at TIMESTAMP NULL

Table 4: ‘Reports table’
report_id (INT, Primary Key, Auto-Increment),
title (VARCHAR, NOT NULL),
description (TEXT),
report_type (VARCHAR),
program (VARCHAR),
total_donations (DECIMAL),
beneficiaries (INT),
report_date (DATE),
created_at (TIMESTAMP DEFAULT CURRENT_TIMESTAMP),
updated_at (TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP)

Table 5: Programs
program_id (INT PRIMARY KEY AUTO_INCREMENT),
name (VARCHAR NOT NULL),
description (TEXT),
target_amount (DECIMAL),
current_amount (DECIMAL, DEFAULT 0),
country (VARCHAR),
status (VARCHAR DEFAULT 'active'),
created_at (TIMESTAMP DEFAULT CURRENT_TIMESTAMP)

API Endpoints (PHP Backend)

`POST /api/register.php`: Accepts user details, hashes password, inserts into `users`.
`POST /api/login.php`: Validates credentials, creates a PHP session.
`POST /api/submit_donation.php`: Writes donation record to the database.
`GET /api/get_profile.php`: Returns logged-in user's details based on active session.
`GET /api/admin_stats.php`: Returns aggregated SUM and COUNT queries for the admin dashboard.